

TIC TAC TOE in C

✖ Problem Statement:

To design and implement a simple **two-player Tic-Tac-Toe game** in C language. The program should allow two users to play alternately, marking their symbols (X or O) on a 3×3 grid. The game should determine the winner when any player aligns three of their symbols horizontally, vertically, or diagonally, or declare a draw when all cells are filled.

🧠 Concepts Used:

- **Loops:** For iterating through the game board and taking multiple turns.
- **Arrays:** To represent the 3×3 board as a 2D array.
- **Functions:** For modular programming (display board, check winner, make move).
- **Conditionals (if-else):** To check win or draw conditions.
- **Switch Statement:** (Optional) To alternate between player turns.
- **Input/Output:** For reading user moves and displaying the board.

📄 Short Description of Working:

1. The game board is represented using a **3×3-character array** initialized with numbers 1–9.
2. Two players take turns choosing a number corresponding to the board position.
3. The program replaces that number with the player's mark (X or O).
4. After each move, the program checks for a winning combination or draw condition.
5. If a player wins, the program displays the winner's name and ends.
6. If all cells are filled without a winner, it displays *"Game Draw!"*.



tictactoe.c:

```

1  /*-----*/
2      Project Title : Tic Tac Toe Game in C
3      Description  : Two-player game played on a 3x3 grid.
4                      Players take turns marking cells as X or O.
5                      The game ends when one player wins or
6                      when all cells are filled (draw).
7      Concepts Used : Arrays, Loops, Functions, Conditionals, I/O
8  /*-----*/
9
10 #include <stdio.h>
11 #include <conio.h> // Optional: for getch() to pause screen
12 #include <stdlib.h> // For system("cls") to clear screen
13
14 // Global 3x3 array to store the game board
15 char board[3][3] = {
16     {'1', '2', '3'},
17     {'4', '5', '6'},
18     {'7', '8', '9'}
19 };
20
21 // Function Prototypes
22 void displayBoard();
23 int checkWinner();
24 int isDraw();
25 void markBoard(int choice, char mark);
26
27 // Main Function
28 int main() {
29     int player = 1; // Player 1 starts first
30     int choice; // Position selected by player
31     char mark; // X or O
32     int status = -1; // -1 = game continues, 1 = win, 0 = draw
33
34     do {
35         system("cls"); // Clear screen (for better visibility)
36
37         printf("\n\tTIC TAC TOE GAME\n");
38         printf("\t-----\n");
39         printf("\nPlayer 1 (X) - Player 2 (O)\n\n");
40
41         // Display current game board
42         displayBoard();
43
44         // Determine which player's turn it is
45         player = (player % 2) ? 1 : 2;
46         mark = (player == 1) ? 'X' : 'O';
47
48         // Ask the player for a move
49         printf("Player %d, enter the position (1-9): ", player);
50         scanf("%d", &choice);
51
52         // Update board with the player's mark
53         markBoard(choice, mark);
54
55         // Check if someone has won or if it's a draw
56         if (checkWinner()) {
57             system("cls");
58             displayBoard();
59             printf("\n==> Player %d wins! \n", player);
60             status = 1; // Game ends with a win
61             break;
62         } else if (isDraw()) {
63             system("cls");
64             displayBoard();
65             printf("\n==> Game Draw! \n");
66             status = 0; // Game ends with a draw
67             break;
68         }
69
70         player++; // Move to next player
71     } while (status == -1);
72
73     printf("\nThank you for playing Tic Tac Toe!\n");
74
75     getch(); // Wait for key press before closing
76     return 0;
77 }
78

```

```

79
80
81  /*-----
82  Function : displayBoard()
83  Purpose  : Displays the current Tic Tac Toe board
84  -----*/
85 void displayBoard() {
86     printf("\n\t %c | %c | %c ", board[0][0], board[0][1], board[0][2]);
87     printf("\n\t---|---|---");
88     printf("\n\t %c | %c | %c ", board[1][0], board[1][1], board[1][2]);
89     printf("\n\t---|---|---");
90     printf("\n\t %c | %c | %c \n\n", board[2][0], board[2][1], board[2][2]);
91 }
92
93  /*-----
94  Function : markBoard()
95  Purpose  : Places player mark on selected position
96  -----*/
97 void markBoard(int choice, char mark) {
98     int row = (choice - 1) / 3; // Convert 1-9 choice into 2D indices
99     int col = (choice - 1) % 3;
100
101     // Check if position is valid and not already taken
102     if (choice < 1 || choice > 9 || board[row][col] == 'X' || board[row][col] == 'O') {
103         printf("Invalid move! Try again.\n");
104         getch();
105     } else {
106         board[row][col] = mark;
107     }
108 }
109
110  /*-----
111  Function : checkWinner()
112  Purpose  : Checks all possible winning combinations
113  -----*/
114 int checkWinner() {
115     // Check rows and columns
116     for (int i = 0; i < 3; i++) {
117         if ((board[i][0] == board[i][1] && board[i][1] == board[i][2]) ||
118             (board[0][i] == board[1][i] && board[1][i] == board[2][i])) {
119             return 1;
120         }
121     }
122
123     // Check diagonals
124     if ((board[0][0] == board[1][1] && board[1][1] == board[2][2]) ||
125         (board[0][2] == board[1][1] && board[1][1] == board[2][0])) {
126         return 1;
127     }
128
129     return 0; // No winner yet
130 }
131
132  /*-----
133  Function : isDraw()
134  Purpose  : Checks if all cells are filled (draw)
135  -----*/
136 int isDraw() {
137     for (int i = 0; i < 3; i++) {
138         for (int j = 0; j < 3; j++) {
139             if (board[i][j] != 'X' && board[i][j] != 'O')
140                 return 0; // Still empty cell, game continues
141         }
142     }
143     return 1; // All cells filled, game is a draw
144 }

```

Output Screenshot (Representation):

```
TIC TAC TOE GAME
-----
Player 1 (X) - Player 2 (O)

  X | 2 | 3
  ---|---|---
  4 | 0 | X
  ---|---|---
  7 | 8 | 9

Player 2, enter the position (1-9):

  X | 0 | X
  ---|---|---
  4 | 0 | X
  ---|---|---
  7 | 0 | 9

==> Player 2 wins!
Thank you for playing Tic Tac Toe!
```

🧠 Logic Flow:

1. Start
2. Initialize board with numbers 1–9.
3. Display board.
4. Take input from player 1 → mark X.
5. Check for winner / draw.
6. If no winner → switch to player 2 → mark O.
7. Repeat steps 4–6 until winner or draw.
8. Display result.
9. End.

💡 Flowchart:

